

Spot–Vol Dynamics: SSR and Smile Transport

Moving the smile when spot moves, without recalibration

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Abstract

When the underlying moves, the implied-volatility smile moves with it, and a desk must know *how* — to risk-manage, to scenario, and to refresh a surface between calibrations. This note documents the Vol-Fitter’s spot–vol dynamics. The organizing quantity is the skew-stickiness ratio $SSR = (d\sigma_{\text{atm}}/d \ln F)/\text{skew}$, the fraction of the skew realized as an ATM-vol change per unit log-spot move. Three canonical regimes fix it: sticky-moneyness ($SSR = 0$, the smile rides with the forward), sticky-strike ($SSR = 1$, vol fixed at each strike), and sticky-local-vol ($SSR \approx 2$ for short-dated skews). We give the analytic transport that realizes any SSR in one line, derive its ATM response, and then the *exact* sticky-local-vol transport (a Hagan-style log-moneyness displacement) and the local-vol-grid relabeling that implements it by simply re-indexing the surface’s strike nodes. Every relabeling path is recalibration-free — a spot move refreshes the surface by arithmetic, not by re-solving; the one exception is the exact sticky-local-vol-grid *reprice*, which pays for its exactness with a single Dupire PDE solve and is chosen explicitly when exactness is wanted over speed. A worked example transports an equity smile under a -5% move in the three regimes.

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1 The question

A calibrated surface is a snapshot at one spot. The next tick moves spot, and the desk needs the new smile *now* — faster than a recalibration, and consistent with a chosen dynamics assumption. The assumptions differ in how much the ATM vol follows the skew as spot moves, and the summary statistic is the *skew-stickiness ratio*

$$\text{SSR} = \frac{d\sigma_{\text{atm}}/d \ln F}{s_0}, \quad s_0 = \left. \frac{\partial \sigma}{\partial k} \right|_{k=0} \quad (\text{the ATM skew}). \quad (1)$$

Heuristic

Read SSR as “how much of the skew is real”. The skew s_0 says that lower strikes have higher vol. When spot drops, does the ATM vol rise to where the old lower-strike vol was (SSR= 1, sticky-strike), not move at all in moneyness terms (SSR= 0, sticky-moneyness), or *overshoot* (SSR $\approx$ 2, sticky-local-vol, the empirically common short-dated regime)? SSR interpolates these, and the transport below realizes any value of it.

2 Conventions, fixed once

Every sign error in smile transport comes from mixing two coordinate systems, so we fix the conventions once, and every formula in this note (and in the code, sign-for-sign) follows them:

Assumption 1 (Sign conventions). (i) The move is $h = \delta = \ln(F_{\text{new}}/F_{\text{old}})$: positive when the forward rises. (ii) Log-moneyness $k = \log(K/F)$ is always measured against the *prevailing* forward — so the same absolute strike has *lower* moneyness after an up-move: $k_{\text{new}} = k_{\text{old}} - h$. (iii) A transported *curve* is a function of new moneyness; a transported *quote* (a fixed strike) is a point whose moneyness label changes.

Convention (ii) is where the classic off-by-sign lives, because it makes curves and quotes move in *opposite* directions in k :

Caution

Under sticky-strike, the vol at each fixed strike is unchanged. To express the *new curve* at new moneyness k , ask which old strike now sits there: the strike whose old moneyness was $k + \delta$. Hence the curve transport reads *forward*, $\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k + \delta)$ — a *plus*. But re-labelling an existing fixed-strike quote goes the other way: its new moneyness is $k - h$ — a *minus*. Both signs are correct, both are in production (the smile transport uses $k + \delta$; the local-vol view re-indexes stored quotes by $k - h$), and swapping either one flips the skew response. The sign-locking tests below are the fence.

3 The analytic transport

Let $h = \delta = \ln(F_{\text{new}}/F_{\text{old}})$ be the log-forward move and $\sigma_{\text{old}}(k)$ the pre-move smile in log-moneyness. The Vol-Fitter transports the smile to new-forward moneyness k by

$$\sigma_{\text{new}}(k) = \sigma_{\text{old}}(k + \delta) + (\text{SSR} - 1) s_0 \delta. \quad (2)$$

The first term is a *sticky-strike re-indexing* (a strike that was at moneyness $k + \delta$ before the move is at moneyness k after it); the second is a uniform *level adjustment* that dials in the desired ATM response while preserving the smile shape. The transport (2) is literally one line (matches production to 10^{-13}):

Listing 1: The SSR smile transport (2) for any regime (distilled from `dynamics/ssr.py`; production's last argument is a `regime` accepting a named regime or a numeric SSR).

```
1 def shifted_smile(k, vol_curve, atm_skew, spot_return, ssr):
2     d = np.log1p(spot_return) # delta = log(
      F_new/F_old)
3     return vol_curve(k + d) + (ssr - 1.0)*atm_skew*d # curve reindex +
      level adj
```

Proposition 1 (ATM response). *Under (2), to first order in δ the ATM vol moves by $\Delta\sigma_{atm} = SSR \cdot s_0 \cdot \delta$, consistent with the SSR definition (1).*

Proof. At $k = 0$, $\sigma_{new}(0) = \sigma_{old}(\delta) + (SSR - 1)s_0\delta = \sigma_{old}(0) + s_0\delta + (SSR - 1)s_0\delta + O(\delta^2) = \sigma_{old}(0) + SSR s_0\delta + O(\delta^2)$. $\square$

So $SSR = 0$ leaves the smile a pure function of moneyness (sticky-moneyness: $\Delta\sigma_{atm} = 0$), $SSR = 1$ cancels the level term and freezes vol at each strike (sticky-strike), and $SSR = 2$ doubles the skew's contribution. Figure 1 transports an equity smile (ATM skew $s_0 = -0.353$) under a -5% move in all three regimes; section 3 shows the linear ATM responses of slope $SSR s_0$.

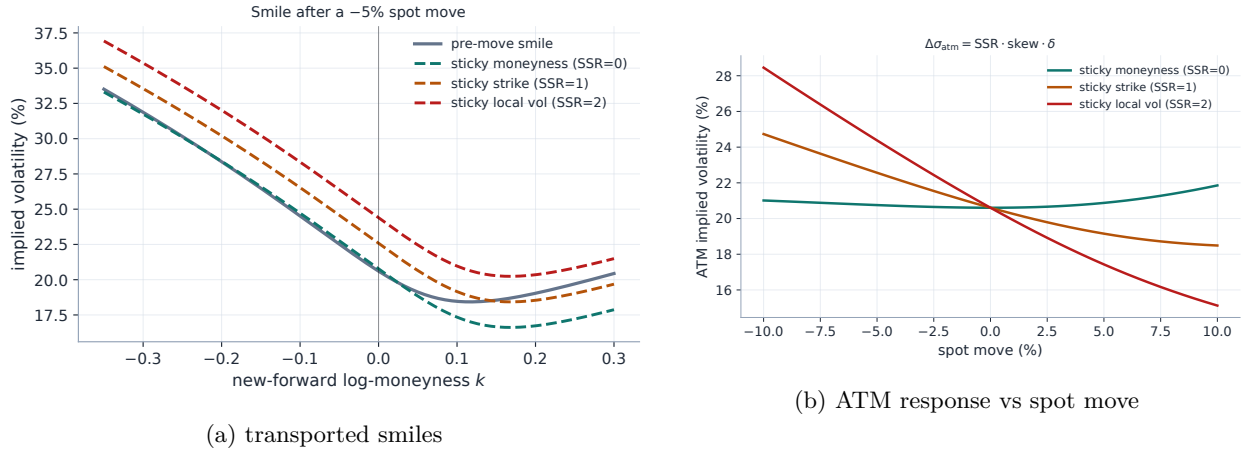


Figure 1: Transporting an equity smile under a -5% spot move in the three canonical regimes. The ATM response is linear with slope $SSR \cdot s_0$.

4 Exact sticky-local-vol

The shift (2) is a first-order, shape-preserving approximation. For the *sticky-local-vol* regime there is an *exact* transport, because holding the local-volatility function fixed in absolute strike determines

the new implied surface completely. In total-variance form, the horizontal-SSR transport is

$$\tilde{w}_1^R(T, k) = w_0(T, k + R h_T), \quad (3)$$

and the exact sticky-local-vol transport replaces the linear displacement by the Hagan log-moneyness map

$$w_1^{\text{LV}}(T, k) = w_0(T, \ell_T(k, h)), \quad \ell_T(k, h) = \log(e^h(e^k + 1) - 1), \quad (4)$$

which is the exact relabeling of moneyness when the local-vol surface is held fixed in absolute strike and the forward moves by h . An optional ATM re-anchor $\sigma_{\text{atm}} \rightarrow \sigma_0 + R \kappa_T h_T$ makes the linear SSR response exact even for large moves.

Expanding the Hagan map for a small move exposes where the famous “ ≈ 2 ” comes from:

$$\ell_T(k, h) = k + (1 + e^{-k}) h + O(h^2), \quad \text{so} \quad \ell_T(0, h) \approx 2h. \quad (5)$$

Holding the local-vol surface fixed makes the new smile read the old curve *two* units of log-moneyness per unit forward move at the money — twice the sticky-strike displacement of δ in (2) — so the ATM vol picks up $2 s_0 h$ to first order: the short-dated SSR ≈ 2 of section 1 is not an empirical accident but the ATM slope of the exact relabeling (4). Away from the money the displacement $(1 + e^{-k})h$ is strike-dependent — larger on the put wing, smaller on the call wing — which is what the linear shift (2) deliberately flattens into a single level adjustment. The expansion is test-locked (traceability table).

4.1 The local-vol-grid relabeling

When the dynamics are driven by the extracted local-vol *grid* (Note 04), the transport is even simpler: it is a *relabeling of the grid’s strike nodes*. A grid node at absolute strike $K_i^0(t)$ moves to

$$K_i^{1,R}(t) = K_i^0(t) e^{(1-R/2)h_t}, \quad \Longleftrightarrow \quad x_i^1 = x_i^0 - \frac{R}{2} h_t, \quad (6)$$

in forward-normalized moneyness x . For $R = 2$ (sticky-local-vol) the factor is $e^0 = 1$: the grid is held fixed in absolute strike and re-priced through the Dupire PDE, which is the *exact* dynamics — here SSR is an *output* of the reprice, not an input, and the value 2 is only its short-maturity limit (5). The exactness is not free: this reprice is the one path in the note that is *not* re-solve-free — it costs a single Dupire PDE solve (Note 04), whereas every relabeling path ((2), (4), (6)) remains pure arithmetic. For $R = 0$ (sticky-moneyness) the grid scales with the forward; $R = 1$ (sticky-strike) sits halfway. This is why the local-vol regime is labeled both `sticky_local_vol` (the analytic Hagan transport) and `sticky_local_vol_grid` (the exact PDE reprice).

In production the local-vol *view* applies all three conventions of assumption 1 at once, per read: each reconstructed smile is wrapped in the curve transport (the $k + \delta$ rule), stored fixed-strike quotes are re-indexed to $k - h$ (the quote rule), and the shared strike-node axis is relabelled by $x \mapsto x e^{-(R/2)h}$ using the longest expiry’s h as the representative move. The three signs are independently test-locked; the grid rule in particular is pinned to the exact triple $(x, x - \frac{1}{2}H, x - H)$ for the three regimes.

Heuristic

The grid relabeling (6) is the cleanest statement of the whole note: a stickiness regime is just a rule for *how the local-vol grid's strikes follow the spot*. $1 - R/2$ is the fraction of the spot move the grid follows — none of it ($R = 2$, grid frozen in strike), all of it ($R = 0$, grid rides the forward), or half ($R = 1$). Everything else — the implied-vol shift (2), the ATM response, the Hagan map — is a consequence of that one rule.

5 What is original here

The SSR and the stickiness regimes are standard (Bergomi, Hagan); the value here is the *unified, recalibration-free* implementation. The single transport (2) realizes *any* SSR (named regime or custom number) by arithmetic on the existing curve; the exact sticky-local-vol path (4)–(6) ties the analytic shift, the Hagan relabeling, and the local-vol-grid reprice into one parameter R ; and because the local-vol regime can be either the analytic transport or the exact PDE reprice, the desk can trade speed for exactness on one slider. No relabeling path ever triggers a re-solve; only the exact sticky-local-vol-grid reprice pays for its exactness with one Dupire solve, and it is chosen explicitly.

6 Traceability

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
ATM moves by $\text{SSR} \cdot s_0 \cdot \delta$; shape preserved	proposition 1	dynamics/ssr.py	test_dynamics.py:: test_atm_vol_moves_by_ssr_times_skew, ::test_shape_is_preserved_up_to_level
$h = 0$ is the identity in every regime	assumption 1	dynamics/transport.py	test_spot_transport.py:: test_h_zero_is_identity_all_regimes
Sticky-moneyness / sticky-strike signs	(2)	dynamics/transport.py	test_spot_transport.py:: test_sticky_moneyness_leaves_smile_in_moneyness_unchanged, ::test_sticky_strike_fixes_vol_at_fixed_strike
Hagan map small-move expansion; LV double-skew response	(4), (5)	dynamics/transport.py::ell_T	test_spot_transport.py:: test_ell_T_small_move_expansion, ::test_sticky_local_vol_double_skew_atm_response
Grid rule pinned to $(x, x - \frac{1}{2}H, x - H)$	(6)	dynamics/transport.py::transport_grid_logk	test_spot_transport.py:: test_grid_node_rule_barycenter, ::test_beta_of_canonical_values

Claim	Object	Code anchor	Test anchor
ATM re-anchor hits the exact linear target	section 4	dynamics/transport.py	test_spot_transport.py:: test_atm_anchor_hits_exact_linear_ssr_target
TransportedSlice wraps implied_w	only section 4	dynamics/transport.py	test_spot_transport.py:: test_transported_slice_matches_function_and_vol
Graph baseline transported under the active regime	Note 14	api/prior_transport.py	test_api_graph.py (SSR= 0 $\Rightarrow$ ATM shift second order)

One honest gap: the local-vol *view*'s composite transport (`api/affine_transport.py`) has no dedicated unit test — its three ingredients (curve rule, quote rule, grid rule) are each locked above, and the composition is covered only at the API level.

A Hyperparameter atlas

Table 2: Spot-vol-dynamics hyperparameters.

Knob	Default	Role
<code>ssr</code>	2.0	SSR value or named regime (<code>sticky_moneyness</code> / <code>sticky_strike</code> / <code>sticky_local_vol</code> / <code>sticky_local_vol_grid</code>).
<i>R</i> regime map	{0, 1, 2, 2}	SSR of the four named regimes.
ATM re-anchor	off	Optional exact linear SSR for large moves ($\sigma_0 + R\kappa_T h_T$).

Performance. The analytic transport (2) and the grid relabeling (6) are $O(\text{grid})$ arithmetic — no optimization, no PDE — so a spot move refreshes the whole surface essentially for free. Only the exact sticky-local-vol-grid reprice costs a Dupire PDE solve, and is chosen explicitly when exactness is wanted over speed. A spot move bumps the per-ticker spot version, so only the moved name's derived grid is re-transported.

References

- [1] L. Bergomi. *Stochastic Volatility Modeling*. CRC Press, 2016.
- [2] P. Hagan, D. Kumar, A. Lesniewski and D. Woodward. Managing smile risk. *Wilmott Magazine*, 2002.
- [3] J. Gatheral. *The Volatility Surface*. Wiley, 2006.